THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE UNIT TEST SET C

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. For $\mathrm{Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu}$, Student A calls it a displacement reaction; Student B calls it double displacement. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

2. Magnesium ribbon burns in air with a dazzling white flame and forms a white powder. The powder is: 1

(a) MgO **(b)** $\mathrm{MgCO_3}$ **(c)** $\mathrm{MgCl_2}$ **(d)** $\mathrm{Mg(OH)_2}$

3. In the electrolysis of water, the volume ratio of hydrogen to oxygen collected is: 1

(a) 1 : 1 **(b)** 2 : 1 **(c)** 3 : 1 **(d)** 1 : 2

4. When zinc reacts with sodium hydroxide, the gas evolved is: 1

(a) carbon dioxide **(b)** hydrogen **(c)** oxygen **(d)** chlorine

5. The process of dissolving an acid in water is: 1

(a) highly exothermic **(b)** endothermic
(c) neither **(d)** a decomposition

6. A solution turns red litmus blue. Its pH is likely to be: 1

(a) 10 **(b)** 4 **(c)** 7 **(d)** 1

7. Silver articles turn black in air due to the formation of: 1

(a) silver nitrate **(b)** silver oxide **(c)** silver chloride **(d)** silver sulphide

8. A student states: “All metals are solids at room temperature.” The correct response is: 1

(a) Correct **(b)** Correct except iron
(c) Incorrect — mercury is a liquid metal **(d)** Cannot be determined

9. Aqua regia is a mixture of conc. HCl and conc. $\mathrm{HNO_3}$ in the ratio: 1

(a) 1 : 3 **(b)** 2 : 1 **(c)** 1 : 1 **(d)** 3 : 1

10. Alkaline $\mathrm{KMnO_4}$ converts ethanol into: 1

(a) ethane **(b)** methanol **(c)** ethanoic acid **(d)** ethene

11. The property of carbon to form long chains by bonding with other carbon atoms is: 1

(a) isomerism **(b)** allotropy **(c)** catenation **(d)** tetravalency

12. Ethanoic acid is commonly known as: 1

(a) formic acid **(b)** lactic acid **(c)** citric acid **(d)** acetic acid

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. Write the balanced equation for the reaction of hydrogen with chlorine and name the type of reaction. 2

14. Why is it advised not to keep curd and sour substances in brass or copper vessels? 2

15. Name a metal and a non-metal that exist as liquids at room temperature. 2

16. What are micelles? 2

17. Write the names of the next two homologues of CH_3OH. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. In the electrolysis of water: 3
(i) Name the gases collected at the cathode and anode.
(ii) What is the ratio of their volumes?
(iii) Why are a few drops of dilute sulphuric acid added?

19. What is water of crystallisation? Write the formula of blue vitriol and explain what happens when it is heated. 3

20. What is roasting and calcination? Give one example of each with equations. 3

21. Write the structural formula of 3
(i) bromopentane
(ii) butanone
(iii) hexanal.

22. How is baking soda prepared? Write the equation and two of its uses. 3

OR

How does the flow of acid rain water into a river make the survival of aquatic life difficult? What is the pH range for human body and aquatic life? 3

23. Explain why ionic compounds 3
(i) conduct electricity in molten state
(ii) have high melting points
(iii) are soluble in water but not in kerosene.

OR

What happens when 3
(i) sodium is dropped in water
(ii) iron filings are added to copper sulphate solution
(iii) copper is heated in air? Write equations.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) Describe an activity to show that acids produce hydrogen gas when they react with metals. 5
(ii) Write the equation for the reaction of zinc with dilute HCl.
(iii) How is the gas tested?

OR

(i) What is meant by strong and weak acids? Give one example of each. 5
(ii) What is a universal indicator?
(iii) Explain the importance of pH in
(a) digestion
(b) soil.

25. (i) Describe the esterification reaction with an equation. 5
(ii) What is saponification? Write its equation.
(iii) Give one use of esters.

OR

(i) What are the two properties of carbon that lead to a large number of carbon compounds? Explain each. 5
(ii) Draw the structure of ethanoic acid.
(iii) Name the functional groups in propanal and propanone.

END OF QUESTION PAPER Total: 50 Marks

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